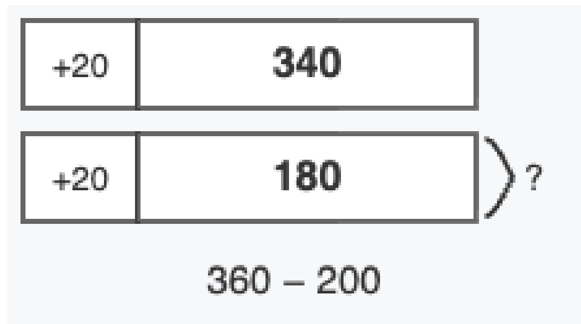
Lesson Objective

Use the associative property to subtract from three-digit numbers.

Materials: linking cubes in two colors (about 20), personal white board

1. Use linking cubes to show that adding the same amount to both numbers keeps the difference the same. Each linking cube is worth 10. Look at the two sticks. What is the difference? Now add one cube to each stick. What is the new difference? Stick A: 7 cubes Stick B: 4 cubes *Answer: 70 − 40 = 30; 80 − 50 = 30. The difference is the same, 30.*	Show students two sticks of linking cubes, one with 7 cubes and one with 4 cubes. Tell students each cube is worth 10, and count them together by tens. Ask students for the number sentence that shows the difference. Record 70 − 40 = 30. Add one cube to the end of each stick. Ask students for the new number sentence and the new difference. Record 80 − 50 = 30. Ask students what changed and what stayed the same. Point out that adding the same amount to both sticks did not change the difference between them.
2. Use compensation with a tape diagram to subtract multiples of ten from three-digit numbers. Solve 340 − 180 by adding the same amount to both numbers to make the problem easier. *Answer: Add 20 to each number. 360 − 200 = 160, so 340 − 180 = 160.*	Draw the tape diagram on the board and label the whole as 340 and one part as 180. Ask students how much they would need to add to 180 to reach 200. Guide students to add 20 to the bottom bar so it shows 200. Ask what they need to add to the top

	bar to keep the difference the same. Add 20 to the top bar so it shows 360. Have students write the new equation, 360 − 200 = 160, and solve. Point out that since the difference did not change, 340 − 180 also equals 160. 
3. Use compensation to subtract a three-digit number close to a benchmark hundred. Solve 425 − 198 by adding the same amount to both numbers to make the problem easier. *Answer: Add 2 to each number. 427 − 200 = 227, so 425 − 198 = 227.*	Ask students how much they need to add to 198 to make a friendly hundred. Then have students work independently to add the same amount to 425, write the new equation, and solve. Check that students added 2 to both numbers, wrote 427 − 200 = 227, and stated that 425 − 198 = 227.

Monitor For	**Instructional Moves**
• Do students count the cubes by tens (10, 20, 30…) in Problem 1, rather than counting each cube as 1?	• If a student counts cubes by ones in Problem 1, point to one cube and remind them each cube is worth 10. Then count the stick together by tens.

• Do students adjust both numbers in the subtraction when applying compensation, not just the subtrahend? • Do students pick a benchmark multiple of 10 or 100 for the subtrahend in Problems 2 and 3? • Do students explain that the difference stays the same because the same amount was added to both numbers?	• If a student adds only to the subtrahend, point to both bars in the tape diagram and ask, "If we add 20 to this bar, what do we need to do to the other bar to keep the difference the same?" • Reinforce choosing a friendly benchmark by asking, "What number close to 198 is easier to subtract?" before students compensate in Problem 3. • To anchor the core idea, after each problem ask students, "Did the difference change? Why not?" and listen for "we added the same amount to both numbers."

Reflection Questions

• When you added one cube to each stick in Problem 1, what happened to the difference? *Answer: It stayed the same, 30.*

• In Problem 2, how did you decide what number to add to 180? *Answer: I added 20 because 180 + 20 = 200, which is a friendly hundred.*

• Why does adding the same amount to both numbers make the problem easier but keep the answer the same? *Answer: Both numbers move up the same amount, so the difference between them does not change. The new problem has a friendly hundred, which is easier to subtract.*

• How can you tell that compensation is a good strategy to use? *Answer: When the number I am subtracting is close to a multiple of 10 or 100, I can add a small amount to both numbers to make an easier problem.*